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Demand Forecasting in Supply Chain Management for Rossmann Stores using Weather Enhanced Deep Learning Model

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ABSTRACT Demand forecasting is one of the essential aspects of supply chain management, as it is linked with the financial performance of the organization. In the retail industry, it is essential to have more accurate forecasts to make suitable decisions. Therefore, the selection of the right forecasting method is considered vital and ideal to meet customer needs. More precisely, this research paper focuses on developing forecasting model for 1115 Rossmann stores located in Europe. Although, previously researchers have been working on developing models to forecast sales demand and to improve accuracy. However, it has been observed that few of the necessary conditions or situations were not being catered for in sales demand forecasting. Such as most researchers used univariate data of total sales for forecasting demand. The internal and external factors such as weather, promotional activity, location of the store, and holidays also play one of the primary roles when it comes to sales demand to forecast. Therefore, it is not specifically a univariate problem but a multivariate problem which have been analyzed in this research. In this research, multivariate dataset including weather variables, other important features have been used in predicting sales demand in supply chain management which helped to achieve better and reliable results. An enhanced deep learning model for sales Demand Forecasting using Weather Data (SDFW) is proposed using Gated Recurrent Unit (GRU) with Grid search. The proposed approach GRU with Grid search showed better performances as compared to previously suggested Long Short Term Memory (LSTM) model. Moreover, Gated Recurrent Unit (GRU) with Grid Search showed significant improvement in sales demand forecasting accuracy when considering weather-related data subsets. These findings will help the Rossmann retail industry in predicting the upcoming sales demand in a more efficient way, which will also optimize their inventory records.

INDEX TERMS sales forecast, supply chain management, retail business, deep learning, grid search, long short-term memory, gated recurrent unit, prediction.

I. INTRODUCTION

Demand forecasting is a crucial task that helps in improving an organizations financial performance [1]. Companies around the world have been taking measures to enhance their market situation in a continuous circle. The business organizations are quite uncertain about many aspects due to which performance is worsening, and as a result, they are having a toll on the financial outlook. According to Forbes, a positive 1% change in the forecast leads to a 2.5% improvement in the amount of inventory needed [2]. Reduced time in delivery, cost, and higher customer satisfaction leads to outstanding results. These outcomes are subjected dramatically on the trust amongst the suppliers throughout the world. Supply Chain is a set of processes used for receiving raw material from suppliers. The raw materials are then processed into final goods and supplied to the consumers via the selling retailers Fig.1. These processes include information compilation, planning, resource management, and performance measurement [2]. The overall organization's business study depends on maximizing the value of the firm by placing the resources at the right place, which has become an essential focus of competitive benefit. It connects the value adding activities to the enterprise's suppliers and customers [3], [4]. Demand predictions can be scaled on a daily or other equal time basis. Most importantly, the items that are produced on a daily basis require daily forecasting. Items that have long shelf life are predicted either weekly or on a monthly basis according to their life. The predictions made on a quarterly or yearly basis helps managers to take decisions in the areas of infrastructure development, planning, and financing to keep the work moving. Demand forecasting can be helpful in the food business as well [5]. In today's sophisticated corporate life, an efficient and effective supply chain is fundamental for success. Many corporations face challenges due to their lack of ability to meet client service expectations regardless of having high levels of slow moving stocks. This is generally what an under performing supply chain leads to. It is the outcome of underprivileged forecasting precision, unproductive planning procedures, and production capacities that are too slow to react to changing market demands. One way to lessen these unpleasant effects of risks is by determining future expectations or sales for goods and services. The employment of demand forecasting is the greatest solution in such cases [6].

FIGURE 1: Life Cycle of Supply Chain

With competition among retailers in the growing market, companies focus more on predictive analytical techniques to lower their costs and increase productivity and profit. The retailers face serious challenges with excessive stock (overstock) and out of stock (stock outs). Excessive stock levels can lead to income loss due to corporate capital bound to stock surplus. Excess inventory can also contribute to higher storage, labour and insurance costs, quality loss, and product type degradation. Out-of-stocking goods can lead to loss of revenue and loyalty to customers. If customers cannot find the products they are looking for, they could move to another competing product or purchase alternative goods. Customer loyalty is very difficult for retailers, especially in low and middle level segments. Sales and loss of customers are a key issue for retailers. Given the competition and financial constraints that the retail industries face, it is very important to have an accurate demand forecasting and retail control system for efficient operations management [6].

The forecast is a projection or an estimation of the real value in a future period. Various modelling techniques for producing forecasts are available. The technology selected varies from the data trend, forecasting horizon, and data quality criteria [7]. After identification of factors, quantitative factors can be estimated, and demand forecasts can be determined using appropriate forecasting techniques [8]. They are based on time-series forecasting approaches while talking about traditional forecasting approaches. These methods help to forecast future demand based further on historical time series data.

In recent years digitization era has changed customer behaviour at a very fast pace. Customers keep on looking for alternatives to the products that are not available at one store or the other. For the companies to manage their products online as well as physically is also becoming challenging day by day. At the same time to keep the organizations competitive with other industrial players, they have to keep monitoring their financial position as well as their reputation in the market to survive. Decision making for goods and services along with SCM contains a complex process of decision making and possess information barriers. Here machine and deep learning techniques are great assets for forecasting in supply chain management [9]. In brief, Machine learning techniques help to predict the exact number of goods and services to be purchased during the given era. Machine based learning techniques can process several significant variables. Traditional demand data estimation is mainly derived from the background of demand, whereas machine based learning techniques employ limitless data sources. This also means that machine learning techniques rely more on data availability [10]. The more the knowledge, the better the learning. A selection of general algorithms in machine learning is used to match demand trends in the entire product portfolio to generate a synced and integrated prediction. However, the sales forecast contains a huge amount of data so the machine learning algorithm does not work well when the amount of data is increased.

According to literature, deep learning approaches surpassed the traditional machine learning algorithms because of scalability of the algorithms, [11], [12]. Deep learning models are a wide set of procedures that use more complex mathematical techniques to select variables. Deep learning models work best for discovering complex and non-linear relationships in data, without needing to make assumptions about external factors. Instead of explicitly weighing variables, many such methods allow us to determine variable importance without worrying about the effects of multicollinearity [13].

Moreover, previous studies have shown that most researchers focused on a univariate data set to predict daily, or other time periods sales demand for selected retail stores sales that include only id, date, and total sales, whereas;

data sets with daily sales can predict daily demand for better management. In the same way, factors such as holidays, promotional activities, weather, and location of the store can also be included in the time series to obtain better and detailed results. Therefore, forecasting of sales demand with a univariate data set is not enough. To get more accuracy and reliable result, a multivariate data set must be considered. also, it has been observed that very few meta-heuristic methods have been used with deep learning algorithms, whereas, the use of other heuristic methods can optimize the algorithm more efficiently to obtain better results. Deep learning can predict improved results for sales demand as compared to traditional techniques.

Obviously, by accurately forecasting future demands, organizations can make informed decisions that help them to grow and prosper. The main problems of demand forecasting in supply chains can be identified as follows.

- Internal and external factors and choice features which impact demand forecasting
- Choice of an accurate forecasting model that can learn complex and non-linear relationships among data
- Refining the forecasting model with the right technique to optimize parameters

To address above issues this research contributes to a retail supply chain demand forecasting framework using weather enhanced deep learning model. Key considerations include: addressing data requirements, applying effective preprocessing and feature selection techniques, implementing deep learning model using Gated Recurrent Unit (GRU) with Grid search for long-term forecasts and model optimization for operational efficiency and enhancing model accuracy by using weather variables. A comparison with previously suggested Long Short Term Memory (LSTM) model with grid search (that is an improved version) is furnished to show the significance of the proposed framework. Finally, Gated Recurrent Unit with grid search was applied to data subsets classified on the basis of weather events due to its better performances.

The paper is organized as follows. Section 2 explains the related work been done regarding the topic in the last few years, a brief comparison of various research papers being used, and the outcome of their research work. Section 3 presents the proposed framework and the and explains the data set, tools and techniques used in this study. In section 4, explains in details all experiments and their results comparison. Section 5 concludes this research.

II. RELATED WORKS

This study focuses on forecasting of the sales demand in supply chain management of the retail industry. Kilimci et al. speaks that demand forecast is one of the biggest problems in the supply chain [14]. Zunic et al. provided the basis for predicting future retail sales accurately and classifying the product portfolio in terms of expected reliability [15]. The proposed approach was based on the Facebook prophet algorithm and a backup test strategy, that can help a retail company greatly. To test the methodology and to demonstrate their capacity within a real-world situation, data collected experimentally in a manufacturing context from one of the largest retail companies in Bosnia and Herzegovina. Taghiyeh et al, proposed a professional method for TS (time series) predicting model, using both comparative and out-of-sample performance collaboratively to train classifiers [16]. Hussain et al. investigated the impact of predictive and comparative studies on SCM demand for products [17]. This study focused mainly on forecasting the demand for future products and increasing the sales turnover of the food retail sector by using two ML algorithms Support Vector Machine (SVM) and Artificial Neural Network (ANN). Kechyn, et al. presented an overall analysis and solution to the basic problem based on TS data [18]. Investigation showed that Convolutional Neural Network (CNN) is good for processing historical data and catching seasonality (yearly pattern), styles, and irregular components. The investigator explained a strategy of using CNN, a cycle of architecture, for sales predictions, which demonstrated to be an effective way of solving TS forecasting problems. However, the CNN architecture needs to be investigated with more layers for more complex tasks. Research on the prediction of seasonal apparel products was conducted [19]. A hybrid model has been proposed by Wang et al. to assess economic sustainability in the development of retail shop products based on a market demand assessment [20]. An experiment was carried out based on the current sales data in retail stores in Guiyang, China. The model was effective in evaluating supply and demand consistency in the retail industry, confirming that it is 92.8 percent accurate. Bandara, et al. says that in e-commerce sector it is important to produce precise and reliable sales forecasts [21]. In situations where large amounts of connected time series are available, it may be beneficial to predict a similar, related time series past behavior by an individual time series.Estimating the revenues before real revenues are recognized to play a key role in the retail sector [22]. Weng, et al. presented supply chain sales forecasting based on a combination model lightGBM and LSTM [23]. Deep learning with long short-term memory networks and random forest in multi-channel market demand prediction was observed [24]. Classical models for statistical revenue modeling such as ARIMA are the most common methods for modeling. However, recent research has shown that deep learning models (e.g. recurrent neural networks) can provide greater predictability in comparison to classical approaches or machine learning models due to their ability to maintain and understand sequential relationships [25]. In recent years, the speed of supply chain demand has been changed rapidly with the development of logistics and e-commerce. The enterprises need to respond to the customer's demand quickly to combine competitive advantages and grasp the market. With the fast development of Big Data and AI, it enables users to accurately forecast long-term sales of goods. The researchers propose the genetic algorithm Gated Recurrent Unit (GA-GRU), a hybrid forecast model that combines the